

Where theory ends: the atlas at a glance

Columns: how much analytic support already exists. Card border = verdict; corner letter = evidence grade.

	Exact closed form known	Asymptotic limit theorem only	Partial special cases only	No theory numerics lead
Anchors	Gaussian-mixture Bayes errorC Gardner storage capacityB			
Symmetry breaking			Multi-index subspace recoveryC Committee specialisation vs sample ratioC	Specialisation under added depthC
Dynamics & landscape		Lazy-to-rich transition vs initialisation scaleB SGD effective temperature vs stationary generalisationB	Double descent vs parameterisation ratioC	Grokking: delayed generalisation vs weight decayC Linear mode connectivity vs widthn/a
Attention & context			In-context learning vs task diversityC Positional vs semantic attentionC	Attention selection vs context lengthC Induction-head formationC
Learning paradigms				Forgetting and transfer vs task similarityC Contrastive dimensional collapse vs temperatureC LoRA rank threshold vs planted perturbation rankC Model collapse vs real-data fractionC Expert specialisation vs load balancingn/a

candidate boundary

transition supported

threshold located (no scale path)

critical response, boundary unresolved

response shift

no resolved response

response shift (boundary off-grid)